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Experimental Investigation of an Ultrathin Manifold Microchannel Heat Sink for Liquid-Cooled Chips

We report an experimental investigation of a novel, high performance ultrathin manifold microchannel heat sink. The heat sink consists of impinging liquid slot-jets on a structured surface fed with liquid coolant by an overlying two-dimensional manifold. We developed a fabrication and packaging procedure to manufacture prototypes by means of standard microprocessing. A closed fluid loop for precise hydrodynamic and thermal characterization of six different test vehicles was built. We studied the influence of the number of manifold systems, the width of the heat transfer microchannels, the volumetric flow rate, and the pumping power on the hydrodynamic and thermal performance of the heat sink. A design with 12.5 manifold systems and 25 μm wide microchannels as the heat transfer structure provided the optimum choice of design parameters. For a volumetric flow rate of 1.3 l/min we demonstrated a total thermal resistance between the maximum heater temperature and fluid inlet temperature of 0.09 $\text{cm}^2 \text{K/W}$ with a pressure drop of 0.22 bar on a $2 \times 2 \text{ cm}^2$ chip. This allows for cooling power densities of more than 700 W/cm^2 for a maximum temperature difference between the chip and the fluid inlet of 65 K. The total height of the heat sink did not exceed 2 mm, and includes a 500 μm thick thermal test chip structured by 300 μm deep microchannels for heat transfer. Furthermore, we discuss the influence of elevated fluid inlet temperatures, allowing possible reuse of the thermal energy, and demonstrate an enhancement of the heat sink cooling efficiency of more than 40% for a temperature rise of 50 K.

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Keywords: manifold, microchannels, impinging jet, heat transfer, electronics cooling

1 Introduction

Recent trends of miniaturization of transistors and of increased integration levels on computer chips enabled a substantial enhancement of the performance of microelectronic devices. However, the increased circuit density and clock frequency lead to an extensive growth of heat dissipation rates of microprocessors, which cannot be removed by traditional air cooling solutions [1]. The current method to address this problem is to reduce the air inlet temperature in datacenters, which causes a significant growth of energy consumption, and thus, a large increase in the operating costs. Brunschwiler et al. [2] showed that a significant reduction in the total heat sink thermal resistance allows for datacenter operation at higher input temperatures above the free cooling limit. This enables the elimination of chillers and, along with this, a reduction of more than 90% of the energy required for cooling. The increased output temperatures of the coolant offer the possibility of energy reuse, which would further increase the efficiency of a datacenter. Furthermore air cooled heat sinks impede the progress of increased integration density as they typically feature large form factors due to a required large surface area and volume for heat transfer. Thinner heat sink structures would provide the opportunity for a denser package of circuit boards, and thus, a reduction in the communication pathlength and of required space for data centers. Hence, there is a strong need for more efficient and compact cooling solutions.

Tuckerman and Pease [3] first introduced the idea of using liquid cooled microchannel heat sinks for cooling electronics. They demonstrated total thermal resistances down to 0.1 $\text{cm}^2 \text{K/W}$ by

implementing 50 μm wide and 300 μm deep microchannels on a 1 cm^2 chip. However, the low thermal resistance was achieved at the expense of large pressure drops across the heat sink. Since then, there have been several studies on different designs for microchannel heat sinks, and a number of reviews are available [4–6]. Improvement to the conventional design can be achieved by introducing hierarchical structured two-dimensional arrays with either bifurcating [7–10] or heterogeneous branching channel networks [11,12] for fluid supply and drain. However, the hydrodynamic resistance of two-dimensional networks results in nonacceptable pressure losses across the heat sink for the required high volumetric flow rates [13]. Harpole and Eninger [14] suggested dividing the heat transfer structure into several zones fed from the third dimension in order to reduce the overall pressure drop of the system. Their idea was studied theoretically, numerically, and experimentally in several investigations [15–20]. Colgan et al. [21] introduced staggered short fins instead of continuous fins for the heat transfer structure. The staggered arrangement provides higher heat transfer coefficients but at the cost of an increased pressure drop. They divided the heat exchange structure of 3 cm^2 into six different zones fed with water or a fluorinated fluid from the top. For a water flow rate of 1.25 l/min they demonstrated an average unit thermal resistance of 0.12 $\text{cm}^2 \text{K/W}$ at pressure drops of about 0.4 bar.

None of the previous studies considered the influence of the manifold on the hydrodynamic and thermal performance of the system. This is an appropriate simplification as long as there are no size constrictions in the third dimension. Brunschwiler et al. [22] considered the influence of the manifold on the performance of the heat sink. They proposed direct single-phase liquid jet impingement on the backside of the chip in combination with a hierarchical tree-like branching manifold system being stacked into the third dimension. The manifold system feeds and drains an array of 40,000 unit cells uniformly distributed on an area of

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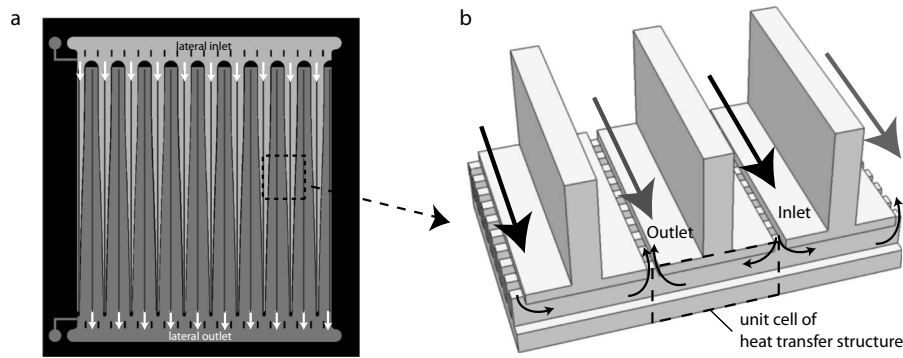


Fig. 1 Schematic of a manifold microchannel heat sink: (a) top view of the manifold with $K=9.5$; (b) isometric section view

4 cm^2 . They demonstrated a minimum unit thermal resistance of $0.17 \text{ K cm}^2/\text{W}$ ($720 \text{ }\mu\text{m}$ chip) at a flow rate of 2.5 l/min and a pressure drop of 0.35 bar . However, with denser packages, there is a demand for thin and efficient heat sinks with only lateral access for fluid supply and drain. In a previous publication [23] we proposed a design for an ultra thin manifold microchannel heat sink. The heat sink consists of impinging liquid slot-jets on a structured surface fed by an overlying two-dimensional manifold system, which requires no access from the third dimension. The proposed design was studied numerically by means of three-dimensional numerical computations and a semi-empirical one-dimensional model. We investigated the basic physical phenomena in the heat transfer structure and the manifold system. We demonstrated for an optimized design a total thermal resistance of $0.087 \text{ cm}^2 \text{ K/W}$ for a volumetric flow rate of 11 l/min and a pressure drop $<0.1 \text{ bar}$.

In this paper we present a thorough experimental characterization of the previously proposed heat sink design concept. We fabricated six different prototypes by means of standard microprocessing. We varied the channel width of the heat transfer microchannel array and the number of manifold systems. A test setup was established, which allows for detailed characterization of the test vehicles. We measured the differential pressure drop across the heat sink and the spatial temperature distribution at the heater surface, and derived the total thermal resistance of the heat sink as a function of the volumetric flow rate. We studied the influence of the microchannel width and the number of manifold systems on the hydrodynamic and thermal performances of the heat sink. The experimental results were compared with the previously proposed semi-empirical one-dimensional model and validated the numerical results qualitatively and quantitatively within an error of 25% . Furthermore, in relation to possible reuse of the generated thermal energy in electronic chips, we discuss the impact of higher fluid inlet temperatures on the performance of the system and demonstrate a cooling efficiency enhancement of the more than 40% for a inlet temperature rise of 50 K .

2 Experimental Methods

2.1 Design and Fabrication. The proposed microchannel heat sink consists of a manifold layer feeding an underlying heat transfer structure of hundreds of parallel microchannel integrated into the backside of the heat dissipating microelectronic device. The microchannel array is divided into thousands of heat transfer unit cells by the overlying manifold (see Fig. 1). The fluid enters the manifold system by a lateral fluid supply and branches into parallel manifold inlet channels. As the fluid flows along the tapered inlet channels it uniformly branches into the underlying microchannels through the slit nozzles located at the bottom surface of the manifold channel. After impinging directly at the backside of the chip, the fluid undergoes a 90° turn and travels along the microchannels orthogonally orientated with respect to the manifold (Fig. 1). While flowing through the heat transfer

channel structure the fluid removes the heat from the electronic components attached below. Subsequently, the fluid leaves the heat transfer microchannel structure upwards through the slit nozzle and merges into the outlet manifold channel, which guides the fluid to a lateral return (Fig. 1). To achieve a uniform cooling capability, the same amount of fluid passes through each unit cell of the microchannel. By introducing tapered inlet and expanding outlet channels a constant pressure gradient across the microchannel array is provided, resulting in a uniform fluid distribution [23].

A test vehicle of the proposed design consists of two silicon dies and a glass chip being all $28 \times 28 \text{ mm}^2$ in size. The microchannel heat sink was fabricated using photolithography and deep reactive ion etching (DRIE) of silicon on 150 mm wafers. One chip contains the heat transfer structure (HT-chip), namely, a $20 \times 20 \text{ mm}^2$ array of parallel microchannels being etched into the center of a $525 \text{ }\mu\text{m}$ thick silicon chip. At the back side of the HT-chip, which is covered by about 70 nm thick silicon oxide layer, we deposited a 300 nm thick NiCr $80/20$ metal layer with a low temperature coefficient of resistance (TCR) of 80 ppm/K being structured by a lift-off resist. The metal layer functions as a resistive heater with a total electrical resistance of $3.1 \text{ }\Omega$ and is of the same size as the microchannel array.

To measure the temperature at the heater plane we embedded two resistive temperature detectors (RTDs), which divided the heater layer into three sections. The RTDs consisted of a 300 nm thick gold layer shaped as a $10 \text{ }\mu\text{m}$ wide and 1.5 mm long serpentine with a total length of 10.3 mm and a mean TCR of 2320 ppm/K . The leads for the four point measurement of the RTD resistance were $100 \text{ }\mu\text{m}$ wide and were directed in pairs to the oppositely arranged contact pads (see Fig. 2(b)). Consequently the heater sections were separated by a $250 \text{ }\mu\text{m}$ gap, providing space for the RTDs. The manifold system and the nozzle array, which connected the manifold and the microchannels, were etched in a double sided DRIE sequence into a 1 mm thick silicon chip (M-chip). The manifold channels were formed first to a depth of $650 \text{ }\mu\text{m}$. In the second step the array of $100 \text{ }\mu\text{m}$ wide slit nozzles were etched $350 \text{ }\mu\text{m}$ deep on the opposite side of the chip. Wider nozzles would be preferable in order to minimize pressure drop [23]. However, the width of $100 \text{ }\mu\text{m}$ was chosen for mechanical and practical reasons, as wider nozzles would collide with the channel width at the end of the tapered manifold inlet channel. The manifold channels were covered by a $500 \text{ }\mu\text{m}$ thick glass chip. The 150 mm glass wafer was coated with a $4\text{--}5 \text{ }\mu\text{m}$ thick polyimide. A leak-proof bond between the silicon wafer containing the M-chips and the glass wafer was achieved in vacuum, by applying a uniform pressure of 7 bar through a membrane to the package backside at 350°C in a membrane oven. The package was cut by a dicing saw into individual chips and bonded to the HT-chips, covered by a $4\text{--}5 \text{ }\mu\text{m}$ thick polyimide, which was deposited on the HT-chip and structured by an O_2 plasma prior to the etching process. We used a flip chip bonder to align the two

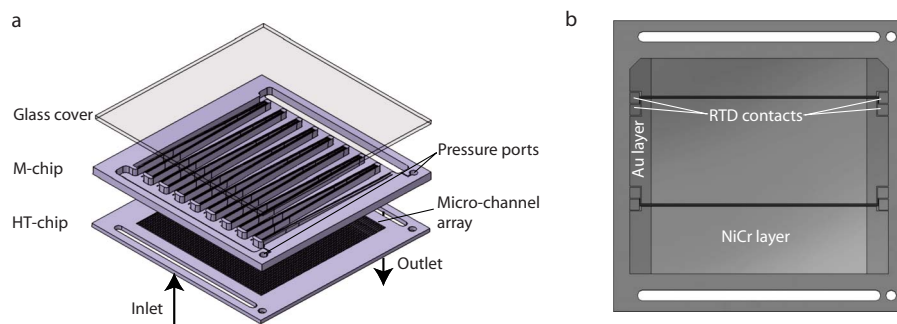


Fig. 2 (a) 3D CAD drawing—iso-metric view on a test vehicle; (b) backside of the HT-chip showing the thin film metal heater with embedded RTDs

chips with an accuracy of $\pm 1 \mu\text{m}$ and prebonded the package at 350°C in a force controlled mode. For the final bond the package was exposed to a pressure of 2 bar at 350°C in a membrane oven. The total height of the heat sink, including a $500 \mu\text{m}$ thick thermal test chip (HT-chip) structured by $300 \mu\text{m}$ deep microchannels, did not exceed 2 mm.

As we pointed out in Ref. [23] the channel width and the number of manifold systems in parallel are the crucial parameters for the performance of the heat sink. Therefore, we fabricated the microchannel array with three different channel widths. The channel depth of the different designs varied slightly as well due to DRIE etch-rate lag, which is a gas diffusion dominated feature size dependent effect. Manifolds with 9.5 and 12.5 systems in parallel were produced. It should be noted that one manifold system was considered as one inlet and one outlet channel, whereas

at the edges just half of a manifold system was created (cf. Fig. 1(a)). In total six different test vehicles were available for thermal and hydrodynamic characterizations (see Table 1).

2.2 Test Setup and Fluid Loop. We designed a test section to interface the individual test vehicles (see Fig. 3(a)). The main body of the test section was fabricated in a machine able glass ceramic due to its low thermal conductivity and mechanical strength. The test vehicles were interfaced by clamping them against a fluorinated rubber gasket bonded to the glass ceramic support by polydimethylsiloxane (PDMS). Thus, the fluid and pressure ports of the test vehicles and the support were tightly connected. At the same time, the electrical contact pads of the test vehicle were contacted by 62 spring-loaded probes. We used eight probes for connecting the RTDs, six probes were employed to

Table 1 Dimensions of the test vehicles

	Chip 1	Chip 2	Chip 3	Chip 4	Chip 5	Chip 6
Channel width, $w_{\text{HT},\text{ch}}$ (μm)	108	107	59	56	26	27
Number of manifold systems, K	9.5	12.5	9.5	12.5	9.5	12.5
Fin thickness, $w_{\text{HT},\text{w}}$ (μm)	92	93	41	44	13	12
Channel height, $h_{\text{HT},\text{ch}}$ (μm)	298	298	264	264	226	226
Nozzle width, $w_{\text{HT},\text{N}}$ (μm)	132	136	136	135	133	133
Manifold wall thickness, $w_{\text{M},\text{w}}$ (μm)	85	85	84	87	85	87
$K=9.5$	$w_{\text{M},\text{ch}}^{\text{in}}(\xi) = w_{\text{M},\text{ch},0}^{\text{in}}(-4 \times 10^8 \xi^5 m^{-5} + 1.22 \times 10^7 \xi^4 m^{-4} + 1.72 \times 10^5 \xi^3 m^{-3} - 8.2 \times 10^3 \xi^2 m^{-2} + 16 \xi m^{-1} + 1)$					
$K=12.5$	$w_{\text{M},\text{ch}}^{\text{in}}(\xi) = w_{\text{M},\text{ch},0}^{\text{in}}(7.2 \times 10^8 \xi^5 m^{-5} - 3.25 \times 10^7 \xi^4 m^{-4} + 6.6 \times 10^5 \xi^3 m^{-3} - 8.14 \times 10^3 \xi^2 m^{-2} + 1.45 \xi m^{-1} + 1)$					

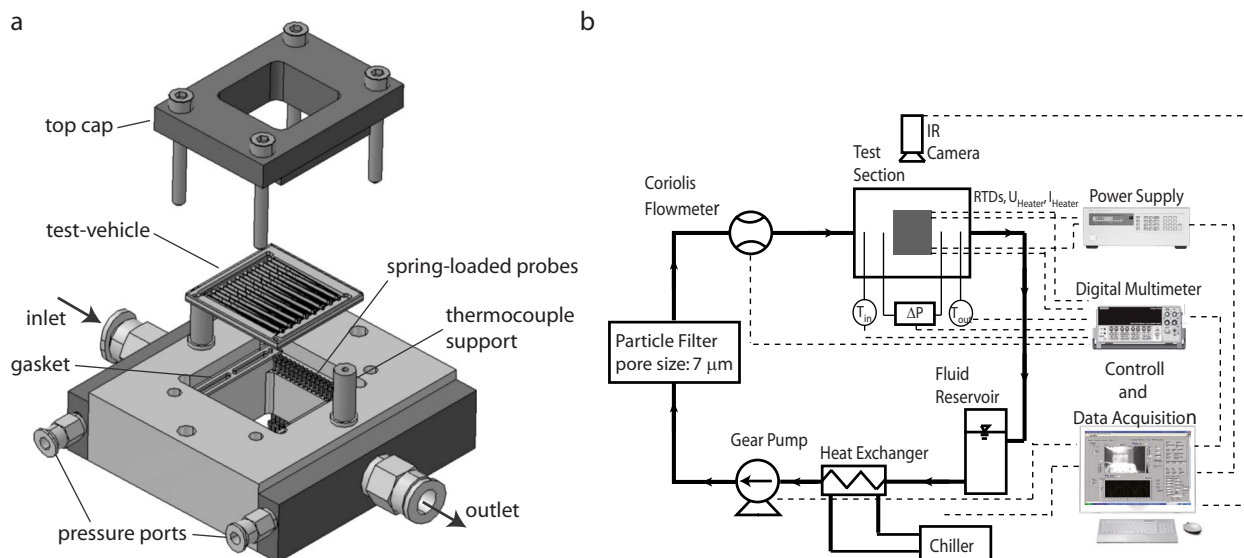


Fig. 3 (a) 3D CAD drawing of the test section; (b) Schematic of the flow loop

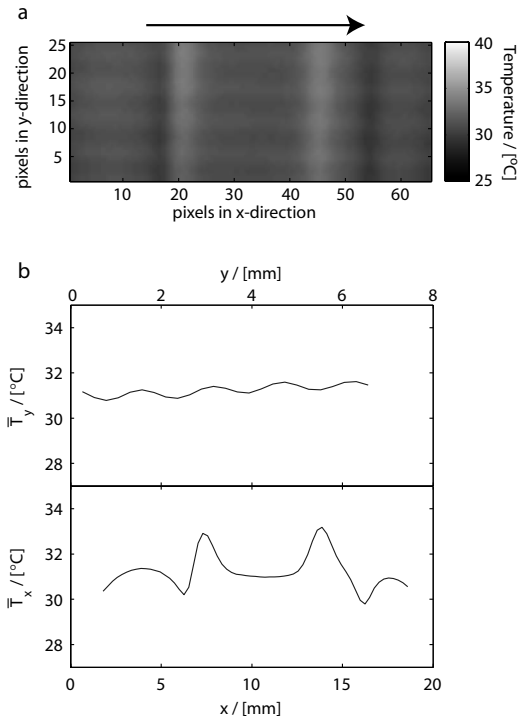


Fig. 4 (a) Spatial temperature distribution of a section of the heater being measured by an IR camera: the arrow indicates the flow direction in the manifold system; (b) top part: temperature distribution in the y -direction being averaged along the x -direction, bottom part: Temperature distribution in the x -direction being averaged along the y -direction

measure the on-chip potential difference across each section of the heater, and the remaining 48 probes provided the electrical connection to the heater, delivering a maximum current of 17 A. Hence, the power dissipation was limited to 850 W. The heat loss from the fluid to ambient was less than 1.5% at typical operation conditions, and the parasitic thermal resistance of the test set up was $20 \text{ cm}^2 \text{ K/W}$.

The test section was integrated into a closed fluid loop. The flow was driven by a magnetic gear pump. We used a Coriolis flow meter to determine the volumetric flow rate with an error of $<3 \text{ ml/min}$. Two T-type thermocouples were integrated into the test section to measure the fluid inlet and outlet temperatures (error $\pm 0.1 \text{ K}$). The pressure drop across the heat sink was quantified by a differential pressure transducer (error $\pm 5 \text{ mbar}$). We measured the on-chip power dissipation by detecting the input current and the on-chip potential difference of each section of the heater. Two RTDs were used to determine the heater plane temperatures at two different locations. We obtained the spatial temperature profile of the heater surface by an infrared camera. A secondary heat exchanger connected to a chiller was used to remove the heat and to control the fluid inlet temperature of $20 \pm 1^\circ\text{C}$. The data reading of the sensors was performed by a digital multimeter and relay switching card, while we recorded the data and operated the system via LABVIEW™.

3 Results and Discussion

In order to characterize the cooling uniformity and to determine the heat sink maximum total thermal resistance, we measured the spatial temperature distribution of the resistive heater by means of an infrared camera. Figure 4(a) shows the temperature distribution of a segment of the heater surface of chip 4 ($w_{\text{HT, ch}} = 50 \text{ } \mu\text{m}$ and $K = 12.5$), operated at a total volumetric flow rate of $\dot{V} = 1 \text{ l/min}$

and a heat input of $q''_{\text{chip}} = 100 \text{ W/cm}^2$. From the infrared image we computed average temperatures in x - and y -directions presented in Fig. 4(b) and defined as

$$\bar{T}_i = \frac{1}{N_j} \sum_{j=1}^{N_j} T_j \quad \text{and} \quad \bar{T}_j = \frac{1}{N_i} \sum_{i=1}^{N_i} T_i \quad (1)$$

where i and j are the pixel counts, and N_i and N_j are the total numbers of pixels in x - and y -directions, respectively. The corresponding flow orientation in the manifold system is indicated by the arrow. The fluid entered the heat sink on the left and branched into parallel manifold inlet channels. From the inlet channel the fluid was distributed to the underlying heat transfer structure through the slit nozzles. When the flow entered the heat transfer structure, fluid impinged directly on the bottom surface of the microchannel, suppressing a development of a hydrodynamic and a thermal boundary layer. Consequently, in this region, high heat transfer coefficients were obtained, resulting in colder heater sections directly underneath the inlet nozzles. This can be seen in the infrared image (cf. Fig. 4(a)) as darker horizontal lines and more quantitative by the local minima in $\bar{T}_y$ (Fig. 4(b)). After impinging on the microchannel bottom surface, the flow underwent a 90 deg turn and traveled along the microchannels, orientated orthogonally to the manifold channels. The fluid heated up along the channel and additionally a hydrodynamic and thermal boundary layer developed, causing a reduction in the local heat transfer coefficient. Toward the end of a unit cell of the microchannel array, the fluid exited through the outlet nozzle, connecting the microchannel to the outlet manifold channel. Thus, a stagnation zone formed opposite to the outlet nozzle, resulting in a decrease in the heat transfer rate and in an increase in the heater temperature toward the outlet (cf. Figs. 4(a) and 4(b)). From the averaged temperature profile in the y -direction we can see that the temperature difference from inlet to outlet was about 0.5 K.

Assuming a uniform fluid distribution along the manifold system we would expect a constant temperature in the x -direction of the heat sink. Except for the spreading effects at the edges of the heat sink, we observed two major deviations from a constant chip temperature along the x -direction. There are two considerable temperature minima at $x = 6.25 \text{ mm}$ and $x = 16.25 \text{ mm}$, and two maxima at $x = 6.66 \text{ mm}$ and $x = 13.33 \text{ mm}$. The positions of the temperature minima correspond to the location of the RTDs. As described earlier, the heater was divided into three sections by the RTDs, causing a strip of $250 \text{ } \mu\text{m}$ in width without any heat input. The lack of heat input caused the temperature drop at the position of the RTDs, which can be clearly seen in the temperature profile.

To understand the temperature maxima, a closer look at the design of the manifold chip is needed. The base plate of the manifold chip was just about $350 \text{ } \mu\text{m}$ in thickness and was perforated with 20 mm long slits representing the inlet and outlet nozzles. To ensure a structural stability and reliability of the manifold chip during the fabrication and packaging process, we decided not to have continuous slits, but to leave uncut $25 \text{ } \mu\text{m}$ wide solid regions at one third and two third of the nozzle length. These regions are functioning as bracings and stiffened the chip. The fact that the locations of the bracings matched well with the positions of the maxima corroborates the assumption that the bracings had an influence on the fluid distribution and consequently on the heat transfer. Thus, we studied numerically their influence on the fluid distribution by a three-dimensional model of a unit cell of the heat sink, as illustrated in Fig. 5(b). The heat sink was represented by an inlet manifold channel, the heat transfer structure, and an outlet manifold channel. Due to the symmetry of the problem we considered half of an inlet and outlet manifold channel. To save computational time the heat transfer structure was considered as a porous media with an anisotropic permeability to account for the impermeable channel walls. The flow field in the manifold domain and the heat transfer structure were determined by solving the three-dimensional continuity and Navier–Stokes equations,

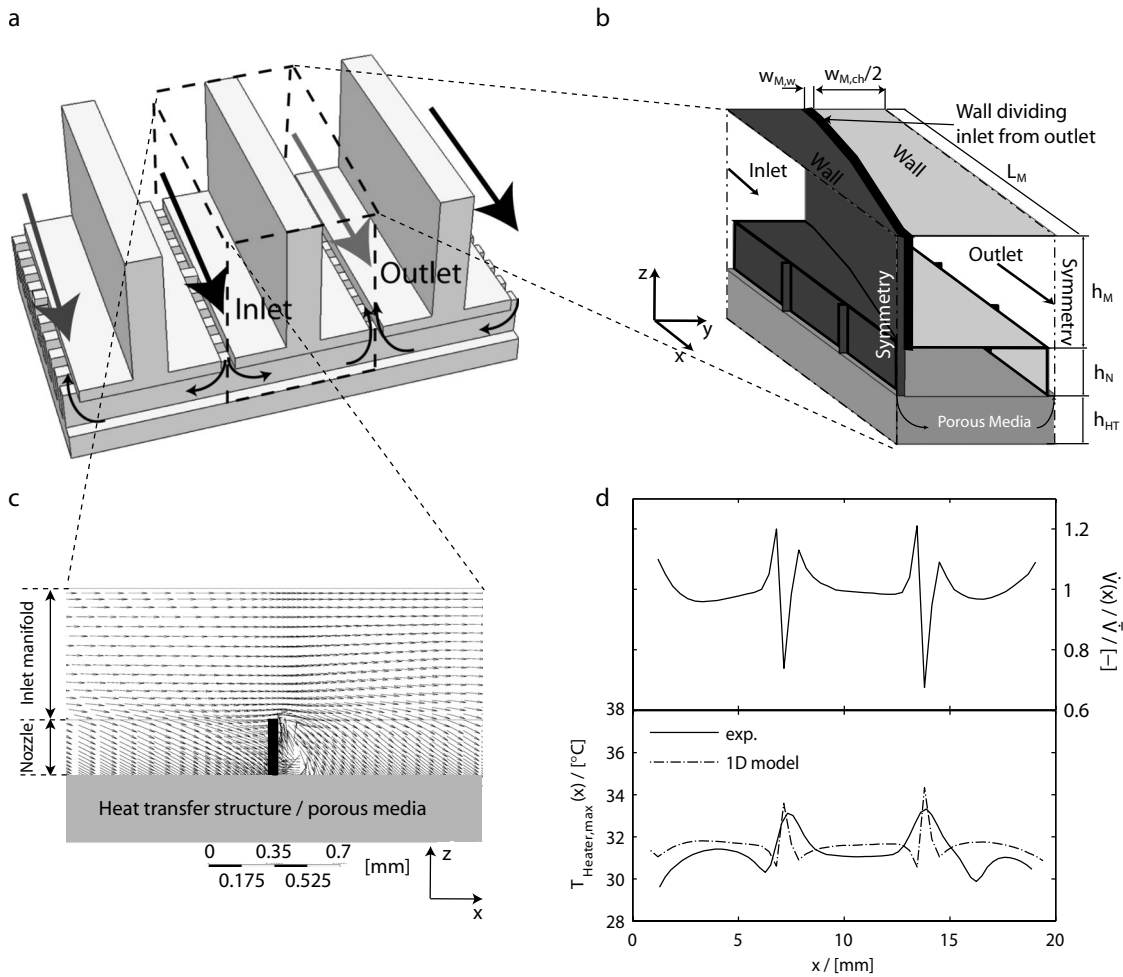


Fig. 5 (a) Isometric section view indicating a unit cell of the heat sink; (b) computational domain of a unit cell of the manifold microchannel heat sink; (c) normalized velocity vectors at the XZ-center plane of the inlet manifold channel around a bracing; and (d) top part: relative volume flux distribution along the manifold system in the porous media, bottom part: comparison of the experimentally and theoretically determined maximum heater temperatures along the manifold system

whereas for the heat transfer domain an additional source term, which was based on the Forchheimer modification of Darcy's law, was introduced to account for the porosity and permeability of the heat transfer structure. At the inlet we applied a constant mass flux boundary condition, and at the outlet an average static pressure of zero was set. All other faces were considered as walls with a no slip boundary condition. The set of partial differential equations with the boundary conditions described above was solved numerically with a finite volume second order high resolution discretization scheme on a structured grid using the solver CFX 11 (ANSYS™). A coupled algebraic multigrid solver is used in conjunction with a convergence criterion for the root mean square mass and momentum residuals of 10^{-5} . The domain was approximated by an anisotropic grid of 611,600 hexagonal cells, whereas the mesh size was verified by a mesh independence study. A more detailed description of the modeling methodology can be found in Ref. [23].

Figure 5(c) shows the normalized velocity vectors at the XZ-center plane of the inlet manifold channel at the position around a bracing for a total volumetric flow rate of 1 l/min. As can be seen from the vector field there is a typical laminar layer flow field in the main channel. At the bottom of the channel the fluid was sucked toward the nozzle into the heat transfer structure and underwent a change in momentum. We observed a substantial distortion of the fluid distribution by the bracing. Upstream of the bracing the flow inside the nozzle was blocked, whereas down-

stream, we observed stalling at the edge of the bracing, causing a recirculation zone behind the bracing inside the nozzle. This recirculation zone blocked a substantial part of the heat transfer structure for fluid supply, resulting in an increased local bulk thermal resistance. To study this phenomenon quantitatively we computed for the local relative volumetric flow rate through 56 planes equally distributed along the center XZ-plane of the heat transfer structure with a width $\Delta\zeta$ of 0.5 mm and the height of the heat transfer structure

$$\dot{V}_i^* = \frac{\int_{s_i - \Delta\zeta/2}^{s_i + \Delta\zeta/2} \int_{\eta=0}^{\eta=h_{HT}} \dot{V}_i d\eta d\zeta}{\frac{1}{N} \sum_{i=1}^N \int_{s_i - \Delta\zeta/2}^{s_i + \Delta\zeta/2} \int_{\eta=0}^{\eta=h_{HT}} \dot{V}_i d\eta d\zeta} \quad (2)$$

As seen from Fig. 5(d), there was a small increase in the volumetric flow rate upstream of a bracing and a sharp drop behind. We used the local fluid distribution to determine the theoretical maximum local heater temperature $T_{x,heater}$ of each of the 56 sections as we considered the change in the bulk thermal resistance due to the fluctuations in fluid distribution

$$T_{heater,max}(x) = R''_{tot}(x) \dot{q}''_{heater} + T_{in} \quad (3)$$

The model to determine the total thermal resistance of the heat sink accounts for a conductive thermal resistance R''_{cnd} of the silicon base, a convection resistance R''_{cnv} , assuming a parallel heat

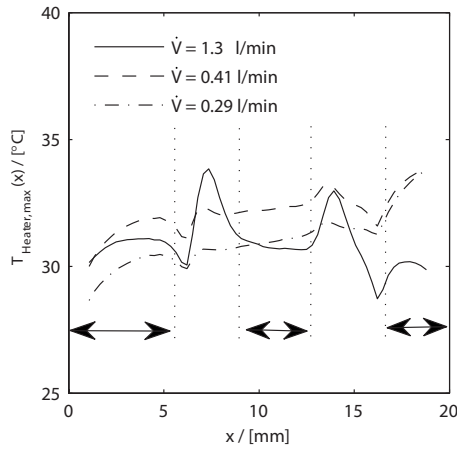


Fig. 6 Local maximum heater temperature for three different flow rates: arrows indicate the range for determination of the absolute maximum heater temperature

flux from the channel base and from the channel wall into the fluid considering a fin efficiency, and a bulk thermal resistance R''_{bulk} due to the finite heat capacity of the fluid

$$R''_{\text{tot}}(x) = R''_{\text{cnv}} + R''_{\text{cnd}} + R''_{\text{bulk}}(\dot{V}(x)) \quad (4)$$

The individual resistances are read as

$$R''_{\text{cnd}} = \frac{h_{\text{HT,ch}}}{\lambda_s}, \quad R''_{\text{cnv}} = \frac{w_{\text{HT,ch}} + w_{\text{HT,w}}}{w_{\text{HT,ch}}\alpha_{\text{cnv}} + 2h_{\text{HT,ch}}\alpha_{\text{cnv}}\gamma_w}, \quad R''_{\text{bulk}} = \frac{L_{\text{ch}}(w_{\text{HT,ch}} + w_{\text{HT,w}})}{c_{p,f}\rho_f\dot{V}_{\text{HT,ch}}} \quad (5)$$

A detailed discussion of the individual components can be found in Ref. [23].

In Fig. 5(d) we compare the experimentally determined local maximum heater temperature in the x -direction to the theoretically expected heater temperature according to Eq. (3) for a heat flux density of 100 W/cm^2 and a volumetric flow rate of 1 l/min . To determine the local maximum heater temperatures, as plotted in Fig. 5(d), we took the averaged heater temperatures in the x -direction and added the maximum deviation of $\bar{T}_y$ from the average temperature. This way we account for the elevated heater temperature at the end of a unit cell of the heat transfer structure due to the heating of the fluid along the channel. Accordingly, the local maximum heater temperature was defined as

$$T_{\text{heater,max}}(x) = \bar{T}_x + |\bar{T}_x - \bar{T}_{y,\text{max}}| \quad (6)$$

Both temperature profiles were in close agreement with the position and the actual temperatures of the maxima. Thus we could demonstrate that the maxima were caused by an induced flow disturbance due to the bracings. In addition this phenomenon points out the strong impact of fluid maldistribution on the thermal performance of the system and emphasizes the need of a thorough design of the manifold and precise fabrication technologies for establishing high performance heat sinks.

Since these bracings would not be needed in a fully automated process, it is acceptable to ignore these maxima. In addition, we did not consider the minima caused by the RTDs. Consequently, we considered the temperature distribution in the ranges being illustrated in Fig. 6 to determine the absolute maximum heater temperature $T_{\text{heater,max}^2}$, which was applied to compute for the total thermal resistance of the system defined as

$$R''_{\text{tot}} = \frac{T_{\text{heater,max}^2} - T_{\text{in}}}{\dot{q}''_{\text{heater}}} \quad (7)$$

We characterized the total thermal resistance of the test vehicles at a constant temperature difference between the maximum heater temperature and fluid inlet temperature of $\Delta T_{\text{max}} = 15 \text{ K}$ and controlled the heater input power. The temperature difference of $\Delta T_{\text{max}} = 15 \text{ K}$ was chosen after a sensitivity study of the total thermal resistance of the system as a function of ΔT_{max} . The total thermal resistance reached an asymptotic value for $\Delta T_{\text{max}} > 11 \text{ K}$, and its error was determined to be less than $\pm 0.005 \text{ cm}^2 \text{ K/W}$. Figure 6 shows the local maximum heater temperature for three different volumetric flow rates. For a flow rate of 1.3 l/min the temperature nonuniformity in the considered range is less than 1 K . If the flow rate is reduced by one third it can be seen that the heater temperature increases at the end of the manifold system. This can be either caused by a change in the fluid distribution due to lower flow rates or by heating up of the fluid along the manifold inlet channel. This phenomenon was more pronounced for a flow rate of 0.29 l/min , resulting in a nonuniformity of about 5 K . As we will demonstrate in the following, at these low flow rates the thermal performance of the heat sink was strongly dominated by its bulk thermal resistance and was thus strongly sensitive to small nonuniformities in the flow rate.

We characterized six test vehicles summarized in Table 1 according to their hydrodynamic and thermal performance. Figure 7(a) shows the pressure drop of all six test vehicles as a function of the volumetric flow rate. The examined range of the volumetric

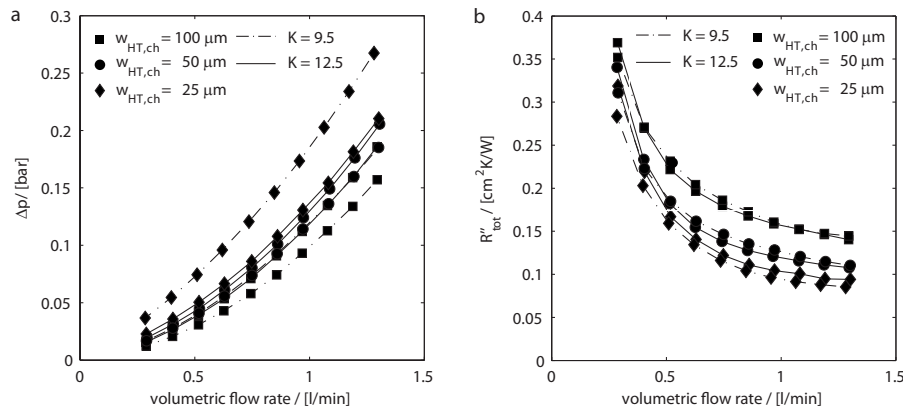


Fig. 7 (a) Total pressure drop across the heat sink and (b) maximum total thermal resistance as a function of the volumetric flow rate for all six test vehicles

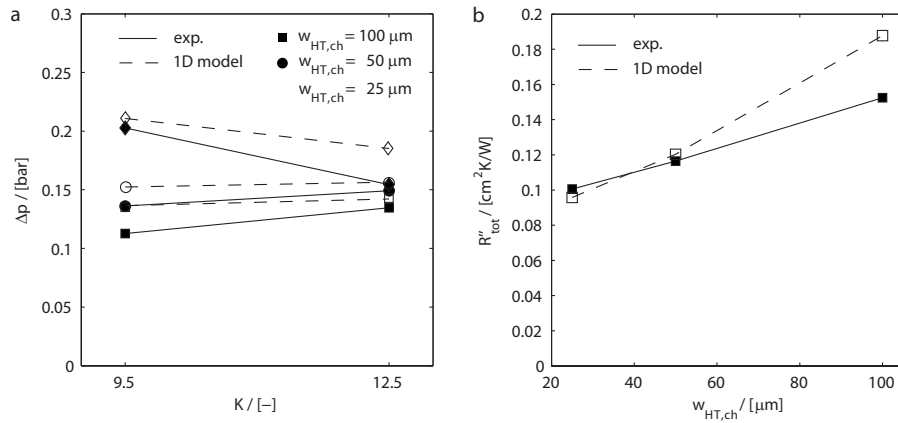


Fig. 8 Comparison of the experimentally and theoretically determined (a) total pressure drop across the heat sink as a function of the number of manifold systems for varying channel width $w_{HT,ch}$ and (b) maximum total thermal resistance as a function of the channel width $w_{HT,ch}$ for $K=9.5$ and a volumetric flow rate of 1.08 l/min

flow rate corresponds to a Reynolds number range at the manifold inlet of $450 < Re_{M,in} < 2800$, where we define the Reynolds number at the inlet as

$$Re_{M,in} = \frac{u_{M,in}(w_{M,in}h_M)^{0.5}}{\nu} \quad (8)$$

Hence we studied the performance of the heat sink from the laminar regime to the border zone of the transition regime between the laminar and turbulent flows in the manifold channel. The Reynolds number for the flow through the heat transfer structure was in the order of unity. Qualitatively for all test vehicles we observed a similar power-law dependence with an exponent being slightly larger than unity between the pressure drop across the heat sink and the volumetric flow rate. We did not observe a rapid change in the pressure drop at a certain volumetric flow rate in any of the test vehicles, indicating that there is no transition from laminar to turbulent flow. The highest hydrodynamic resistance was measured for chip 5 ($w_{HT,ch}=25 \mu m$ and $K=9.5$), whereas chip 2 ($w_{HT,ch}=100 \mu m$ and $K=12.5$) provided the smallest hydrodynamic resistance. The total hydrodynamic resistance depends on both the number of manifold systems K and the width of the heat transferring microchannels $w_{HT,ch}$.

To discuss this dependency more in-depth we present in Fig. 8(a) the pressure drop of all six test vehicles for a constant volumetric flow rate of 1.08 l/min. The test vehicles with $w_{HT,ch}=100 \mu m$ exhibit a smaller pressure drop for $K=9.5$ manifold systems than for $K=12.5$. The total pressure drop across the heat sink is composed of the pressure losses across the manifold system and across the heat transfer structure

$$\Delta p_{tot} = \Delta p_M + \Delta p_{HT} \quad (9)$$

where the individual terms are read as

$$\Delta p_{HT} = 4f_{app,N} \frac{\rho_f \mu_N^2 L_N}{d_N} + \frac{2f_{app,ch} \rho_f \mu_{ch}^2 L_{ch}}{d_{ch}} + 0.4 \frac{\rho_f \mu_N^2}{2} + \frac{\rho_f \mu_{ch}^2}{2} \quad (10)$$

and

$$\Delta p_M = \int_0^{L_M} - \frac{2f_{app}(\xi_M, \bar{u}_M) \rho_f \bar{u}_M^2(\xi_M)}{d_M(\xi_M)} d\xi_M \quad (11)$$

A detailed discussion of the equations can be found in Ref. [23].

If we increase the number of manifold systems, the initial cross section of a manifold channel narrows as the width of the manifold channel is reduced according to

$$w_{M,ch} = \frac{L_{heater}}{2K} - w_{M,w} \quad (12)$$

where $w_{M,w}$ is the minimum wall thickness required for sufficient structural stability. The pressure drop across a rectangular straight channel is proportional to the volumetric flow rate and drops with the fourth power of the hydraulic diameter. Following Muzychka and Yovanovich [24] we defined the hydraulic diameter d_h as the square root of the channel cross section. Assuming that the flow branched uniformly into the parallel manifold systems, the volumetric flow rate through each manifold system is read as

$$\dot{V}_M = \frac{\dot{V}_{tot}}{K} \quad (13)$$

Hence along with Eqs. (12) and (13) the pressure drop across the manifold was directly proportional to the number of manifold systems K and can be written as

$$\Delta p \sim \frac{C_1 \dot{V}_{tot}}{d_h^4} \quad \text{with } d_h = (hw)^{0.5} \Rightarrow \Delta p_M \sim C_2 \dot{V}_{tot} K \quad (14)$$

On the other hand increasing the number of manifold systems reduces the fluid path through the heat transfer structure, and thus, its hydrodynamic resistance. Thus an increasing total pressure drop across the heat sink with increasing number of manifold systems demonstrates that the rise of the hydrodynamic resistance of the manifold systems outweighed the reduction in the pressure loss across the heat transfer structure. If the channel width of the heat transfer structure $w_{HT,ch}$ is reduced, we increase its hydrodynamic resistance, while keeping the pressure loss across the manifold system unaffected. Thus, below a certain channel width $w_{HT,ch}$, the pressure loss across the heat transfer structure will dominate the overall pressure drop of the heat sink. Consequently, in this range, a reduction in the hydrodynamic resistance of the heat transfer structure due to an increased number of manifold systems from $K=9.5$ to $K=12.5$ would outweigh the accompanied rise of pressure loss across the manifold. As seen from Fig. 8(a) for a channel width of $w_{HT,ch}=50 \mu m$ less manifold systems ($K=9.5$) were beneficial, whereas for a channel width of $w_{HT,ch}=25 \mu m$ it was favorable to increase the number of manifold systems to $K=12.5$. Hence, for the present parameter set, the transition formed a manifold dominated to a heat transfer structure dominated hydrodynamic resistance, which occurred between $w_{HT,ch}=50 \mu m$ and $25 \mu m$.

In a previous paper [23] we presented a semi-empirical one-dimensional model developed based on the results of a three-dimensional numerical model. It considers the pressure drop across the manifold, nozzles, and microchannels. To demonstrate its accuracy, we applied this model to the present test vehicles and used the geometric parameters, as summarized in Table 1 as input parameters. In Fig. 8(a) we compare the pressure losses of all six test vehicles estimated by the semi-empirical one-dimensional model to the experimental results. The model represents the hydrodynamic performance qualitatively and quantitatively with a maximum deviation of less than 25%.

The total thermal resistance of the six test vehicles as a function of the volumetric flow rate is presented in Fig. 7(b). We observed a power-law decay of the total thermal resistance for an increased volumetric flow rate. As shown in Ref. [23] this power-law decay was caused by a reduction in the bulk thermal resistance of the heat sink. The test vehicle chips 5 and 6 with a channel width of the heat transfer structure of $w_{HT, ch}=25 \mu m$ provided the smallest total thermal resistance for a given volumetric low rate. Increasing the channel width $w_{HT, ch}$ results in higher total thermal resistances, as shown in Fig. 7(b).

The number of manifold systems K had no significant influence on the thermal performance of the heat sinks. For a constant total volumetric flow rate a reduction in the number of manifold systems causes an increase in the local volumetric flow rate through one unit cell of the heat transfer structure. We demonstrated numerically in Ref. [23] that the convective thermal resistance is almost unaffected by an increase in the local volumetric flow rate, hence, by an increase in velocity. In addition, the bulk thermal resistance of the system remains the same as the fluid path length through the heat transfer structure decays proportional to the number of manifold systems and compensates the increased local volumetric flow rate. Hence, we would not expect a dependency of the total thermal resistance on the number of manifold systems, which is consistent with the experiment (cf. Fig. 7(b)).

Figure 8(b) illustrates the total thermal resistance of the heat sink as a function of the channel width of the heat transfer structure for a constant volumetric flow rate of 1.08 l/min and a number of manifold systems $K=9.5$. The total thermal resistance increases linearly with increasing channel width. In addition, we plotted the total thermal resistance being computed by means of the semi-empirical one-dimensional model (cf. Eq. (4)). The model shows qualitatively the same dependency between the total thermal resistance and the width of the channels of the heat transfer structure. However, there is a discrepancy in the slope of the functional relationship, which becomes obvious especially for the larger channel width of $w_{HT, ch}=100 \mu m$. The semi-empirical one-dimensional model does not explicitly account for any heat transfer from the nozzle walls into the fluid. The test vehicles with a channel width of $w_{HT, ch}=100 \mu m$ had aspect ratios of about three, providing a good thermal connection through the channel walls to the nozzle section (fin efficiency of 86%). This might be the reason for underestimating the thermal performance of the system especially for larger channel widths. Nevertheless, the maximum quantitative deviation is less than 25%.

In the following, we compare the efficiency of the different heat sink designs. The efficiency of a heat sink is defined by the ratio of the removed heat flux per invested pumping power, also known as its coefficient of performance (COP) [25]. This scales reversely with the total thermal resistance of the system. Hence, a lower total thermal resistance at a constant pumping power demonstrates a higher heat sink performance. In Fig. 9 we compare the total thermal resistance of the different designs with a number of manifold channels of $K=9.5$ as a function of an idealized pumping power P , which is defined as

$$P = \Delta p \dot{V}_{tot} \quad (15)$$

In the case of high pumping powers and high volumetric flow rates the design of the heat transfer structure with the smallest

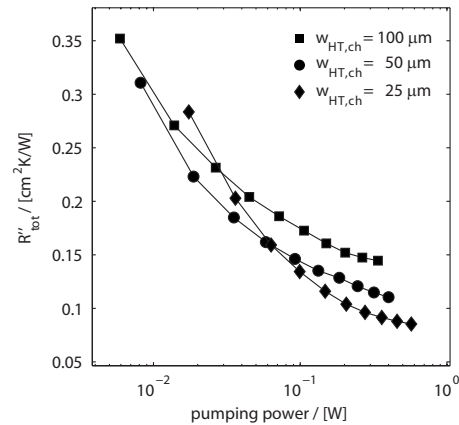


Fig. 9 Maximum total thermal resistance as a function of the pumping power for varying channel width $w_{HT, ch}$ and $K=9.5$

channel width, namely, $w_{HT, ch}=25 \mu m$, performed most efficiently, although it showed the highest hydrodynamic resistance (cf. Fig. 7(a)). At a constant pumping power the volumetric flow rate was lower for the test vehicle with $w_{HT, ch}=25 \mu m$ than for the other two designs with larger channels, and consequently, it resulted in a higher bulk thermal resistance. However, due to the lower convective thermal resistance of this test vehicle, the available thermal mass flux was used more efficiently resulting in higher outlet temperatures for a constant heat flux.

If the pumping power is reduced, the increase in the total thermal resistance of the test vehicle with $w_{HT, ch}=25 \mu m$ was more pronounced than for the other two devices. Since the convective thermal resistance was lower than the one of the other devices, the transition at which the bulk thermal resistance became the dominant part in the resistance chain occurred at higher volumetric flow rates, and consequently, at higher pumping powers. Thus the total thermal resistance of the test vehicle with $w_{HT, ch}=25 \mu m$ showed a stronger dependence on the pumping power. For pumping powers below 0.06 W it would be beneficial to increase the channel width of the heat transfer structure from $w_{HT, ch}=25 \mu m$ to $50 \mu m$ in order to reduce the hydrodynamic resistance, and therefore, the bulk thermal resistance. At this point the increase in the convective thermal resistance due to an enlargement of the channel width would be compensated by the reduction in the bulk thermal resistance.

Next, we demonstrate the potential of the heat sink design to be used in combination with elevated inlet temperatures to reduce the datacenter energy costs and to make energy reuse possible. We studied the influence of increased fluid inlet temperatures on the performance of the best performing test vehicle with a channel width of $w_{HT, ch}=25 \mu m$ and a number of manifold systems of $K=12.5$. Figure 10(a) depicts the pressure drop and the total thermal resistance of this device as a function of the fluid inlet temperature at a constant volumetric flow rate of 1 l/min and a constant heat flux density of $100 W/cm^2$. An increase in the fluid inlet temperature from $20^\circ C$ to $70^\circ C$ lowered the pressure drop across the heat sink by about 40% due to a 60% reduction in the dynamic viscosity of water. The pressure drop reduction was not linear as the dynamic viscosity of water decays exponentially with temperature.

On the other hand, the total thermal resistance of the heat sink was increased by about 12%, mainly attributed to a 20% reduction in the silicon thermal conductivity. A reduction in the thermal conductivity of silicon contributed to an increase in the conduction thermal resistance of the silicon base, causing a 6% increase in the total thermal resistance. In addition, the fin efficiency was decreased, causing a reduction in the convective thermal resistance. Simultaneously, the fluid thermal conductivity was increased by about 10%, resulting in an increased convective heat

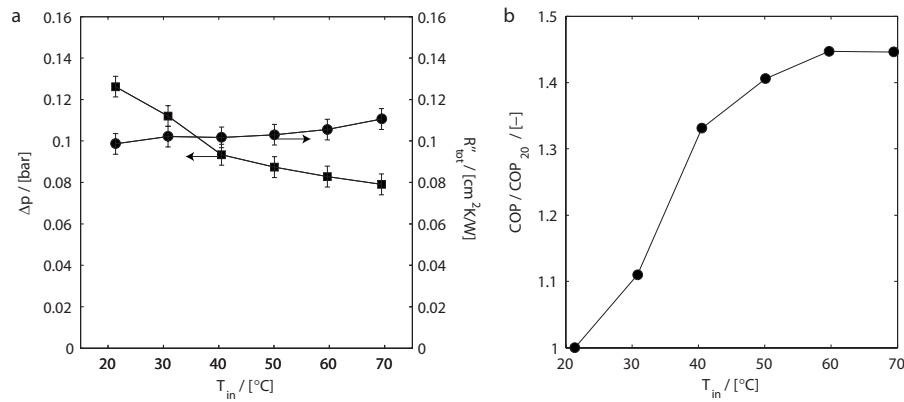


Fig. 10 (a) Total pressure drop (squares) and maximum total thermal resistance (circles) and (b) relative COP as a function of the fluid inlet temperature for $w_{HT,ch}=25 \mu\text{m}$, $K=12.5$, $\dot{q}''_{heater}=100 \text{ W/cm}^2$, and $\dot{V}=1 \text{ l/min}$

transfer. However, the total thermal resistance of the heat sink was only slightly affected by the convective thermal resistance as it accounts for only a share of 20% of the chain of thermal resistances. The volumetric heat capacity of the fluid decreased by about 2%, which directly affected the bulk thermal resistance of the system. To evaluate the increase in efficiency of the heat sink due to elevated inlet temperatures, we determined the relative COP, normalized to the COP at 20 °C. Hence, we write the relative coefficient of performance for a constant maximum temperature difference between heater and fluid inlet as

$$\frac{COP}{COP_{20}} = \frac{\dot{q}_{heater}}{P} \frac{P_{20}}{\dot{q}_{heater,20}} = \frac{P_{20} R''_{tot,20} \Delta T_{max}}{P R''_{tot} \Delta T_{max,20}} = \frac{P_{20} R''_{tot,20}}{P R''_{tot}} \quad (16)$$

Figure 10(b) depicts the relative enhancement in the COP as a function of the fluid inlet temperature. Due to the nonlinear behavior of the pressure drop as a function of the inlet temperature (cf. Fig. 10(a)) we observed a saturation curve for the COP. At inlet temperatures above 60 °C, the pumping power reduction and the increase in the total thermal resistance were balanced. Hence, a further rise of the inlet temperature did not influence the efficiency of the heat sink. Concluding, the performance of the heat sink could be enhanced by more than 40% for a 50 K rise of the fluid inlet temperature (compared with an inlet temperature of 20 °C), resulting in a maximum heater temperature of 81 °C for a fluid inlet temperature of 70 °C.

4 Conclusion

A novel, ultrathin manifold microchannel heat sink was fabricated and investigated experimentally. We processed six different prototypes with varying channel widths of the heat transfer structure and different numbers of manifold systems. A test loop for accurate hydrodynamic and thermal characterizations of the test vehicles was established. We studied experimentally the sensitivity of the heat sink hydrodynamic and thermal performance on the width of the microchannels, the number of manifold systems, the volumetric flow rate, and the pumping power. The present experimental results validated our previous semi-empirical one-dimensional model [23] qualitatively and quantitatively within a maximum deviation of 25%.

It was demonstrated that the optimum design of the heat sink was determined by the interaction between the number of manifolds and the channel width of the heat transfer structure. The optimum was shifted to wider heat transfer channels for lower pumping powers. The best performing prototype ($w_{HT,ch}=25$ and $K=12.5$) closest to the optimum choice of design parameters provides a total thermal resistance of $0.09 \text{ cm}^2 \text{ K/W}$ at a volumetric flow rate of 1.3 l/min and a pressure drop of 0.22 bar for a four square centimeter chip area and an inlet temperature of 20 °C.

This allows for cooling power densities of more than 700 W/cm^2 for a maximum temperature difference between the fluid inlet and maximum chip temperature of 65 K.

Typical power densities of high performance microprocessors are currently in the range of 100 W/cm^2 . Our current optimum design ($w_{HT,ch}=25 \mu\text{m}$ and $K=12.5$) allows for elevated inlet temperatures of up to 70 °C, resulting in a maximum chip temperature of 81 °C for a flow rate of 1 l/min. The low thermal resistance and the tolerable high inlet temperatures provide the possibility of free cooling in all climate zones, eliminating the need for chillers. In addition, the rise of the inlet temperature to 70 °C enhances the overall efficiency of the heat sink by more than 40%. Hence, not only the possibility of energy reuse is provided by increasing the inlet temperatures, but the overall performance of the cooling solution is enhanced by a reduction in the total pressure drop across the heat sink.

The current performance of the heat sink could be further improved by reducing the nozzle length and by increasing the manifold channel height. This would drastically lower the hydrodynamic resistance of the heat sink [23]. A reduction in the hydrodynamic resistance of the manifold would shift the design optimum toward higher numbers of manifold systems. Hence, the flow path through the heat transfer structure would be reduced, which allows for narrower channel widths of the heat transfer structure and accordingly entails a reduced total thermal resistance of the heat sink.

If there is a demand for other liquid coolants than water, the optimum design of the heat sink will change. For example, the operation with fluids having lower heat capacity than water would shift the optimum to larger heat transfer channels and less manifold systems in parallel in order to reduce the hydrodynamic resistance of the system and therewith increase the volumetric flow rate to compensate the increased bulk thermal resistance. Hence the decoupling of the fluid distribution from the intrinsic heat transfer provides the advantage of damping effects from, e.g., changed fluid properties.

The proposed design implies that the heat transfer structure, namely, the microchannel array, is incorporated directly in the back side of the microelectronic device. If a patterning of the microelectronic device was not desired, the heat transfer structure could be fabricated in a separate silicon chip, which could be bonded to the microelectronic device by a eutectic AuSn 80/20 thin film solder, as proposed by Brunschweiler et al. [26]. The AuSn provides a bulk thermal conductivity of 57 W/m K at a bondline thickness of about $5 \mu\text{m}$, resulting in an additional conductive thermal resistance of less than $0.001 \text{ cm}^2 \text{ K/W}$. Therefore, the thermal resistance of the bond is almost negligible, and

the total thermal resistance of the cooling solution can be determined by accounting for the additional conduction resistance of the microelectronic device.

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Notation

c	= constant
c_p	= heat capacity (J/kg K)
COP	= coefficient of performance
d	= characteristic diameter (m)
f	= friction factor
h	= height (m)
K	= number of parallel manifold systems
L	= length (m)
N	= quantity indicator
p	= pressure (Pa)
P	= pumping power (W)
$\dot{q}$	= heat flux (W)
R	= thermal resistance (K/W ⁻¹)
T	= temperature (K)
$\dot{V}$	= volumetric flow rate (m ³ s ⁻¹)
w	= width (m)
x, y, z	= global coordinates (m)

Greek Letters

α	= heat transfer coefficient (W/m ² K)
γ	= fin efficiency
λ	= thermal conductivity (W/m K)
ρ	= density (kg/m ³)
ξ	= local coordinate in the streamwise direction (m)
ζ, η	= local coordinate in the transverse direction (m)

Subscript/Superscript

"	= area normalized (m ⁻²)
i, j	= indices
ch	= channel
cnd	= conduction
cnv	= convection
f	= fluid
h	= hydraulic
HT	= heat transfer structure
in	= inlet
M	= manifold
max	= maximum
N	= nozzle
s	= solid
tot	= total
w	= wall

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